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Impute-VSS: A comprehensive web-based visualization and simulation suite for comparative data imputation and statistical evaluation

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ABSTRACT

In today's technological landscape, where data processing forms the backbone of modeling and predictive analytics, data imputation is crucial in filling missing values within datasets using statistical techniques. However, a notable gap exists in the literature for a toolkit that intuitively offers data imputation. This research aims to bridge the gap by introducing Impute-VSS. This web-based imputation suite offers comparative simulations along with a novel imputation management pipeline for in-depth interpretation of imputation techniques. This manuscript showcases the technical premise of Impute-VSS and emphasises its potential as an inclusive data imputation suite for practitioners and researchers.

Metadata

Nr	Code metadata description	*Please fill in this column*
C1	Current code version	*v1.0.0*
C2	Permanent link to code/repository used for this code version	https://github.com/vartul-shrivastava/Impute-VSS-26022025
C3	Permanent link to reproducible capsule	-
C4	Legal code license	*MIT*
C5	Code versioning system used	*GitHub Desktop*
C6	Software code languages, tools and services used	*Python 3.12.3, Numpy, Pandas, Flask, HTML, CSS, JavaScript, Chart.JS*
C7	Compilation requirements, operating environments and dependencies	*Python 3.12.3 or above, Modern web Browser, Flask-support in Python environment*
C8	If available, link to developer documentation/manual	https://github.com/vartul-shrivastava/Impute-VSS-26022025/blob/main/README.md
C9	Support email for questions	vartul.shrivastava@gmail.com, shekhars@iimidr.ac.in

1. Motivation and significance

In this data-driven world, where most of the decisions and actions are influenced by information-backed premises. Datasets have emerged as a crucial resource, which helps in extracting patterns and predicting trends that formulate insights [1]. In these datasets, missing values commonly exist across the domains, which could be due to a variety of known and unknown reasons such as human error, and data-gathering problems, among many others [2,3]. Such missing values are concerning, as they can hamper analysis which might render inconsistent reasoning. To address this missing data problem, which can hinder the results of modeling, analytics and decision-making processes; Data Imputation has become an indispensable part of data processing workflows [4–6]. Data Imputation is a prevalent methodological procedure, that aids in the prediction of missing values using statistical techniques [7]. In the current literature offering, many toolkits and packages exist that facilitate imputation, which notably include Alteryx [8], AllInOne Pre-Processing [9], MDI Toolbox [10] and many others. These aforementioned toolkits and many others have helped in imputing datasets across the fields. However, an imputation toolkit that leverages comparative assessments through visuals and modular workflows remains scarce (refer to Table 1 for comparative analysis).

Compared to existing toolkits that often rely on only a few evaluation metrics, Impute-VSS offers a more comprehensive approach by incorporating a diverse range of imputation techniques. The toolkit employs advanced machine learning algorithms—such as Gradient Boosting, Random Forest Regressor, and Support Vector Regression—alongside conventional methods like mean, median, and mode imputation, random imputation, complete case analysis (CCA), K-Nearest Neighbors (KNN), and Multiple Imputation using Chained Equations (MICE). These

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Table 1
Comparative analysis of Impute-VSS with other prominent Imputation Toolkits.

Parameter	Impute-VSS	AllInOne Pre-Processing	Alteryx	MDI Toolbox
Comparative Imputation Assessments	✔	×	×	×
Univariate Imputation	✔	✔	✔	✔
Multivariate Imputation	✔	✔	×	✔
Imputation Pipeline support	✔	×	×	×
GUI-based front-end	✔	✔	✔	✔

methods coupled with comprehensive distributional metrics (e.g., KDE overlap, skewness, kurtosis, KS statistics, and KL divergence), help in assessing imputation quality. Furthermore, Impute-VSS offers interactive visualizations and a modular imputation pipeline management system, features that are scarce in other available toolkits, thereby providing a more holistic and user-friendly platform for both researchers and practitioners.Thereby, aimed to position itself as an inclusive suite for data imputation purposes.

In this study, we have developed Impute-VSS, a comprehensive imputation suite that offers a web-based user interface along with novel features, such as comparative assessment and statistical evaluation of various univariate and multivariate imputation methods along with machine learning algorithms on dataset variables, and an imputation pipeline to manage the steps undertaken during the analysis.

This research undertaking aims to showcase the technicalities of Impute-VSS along with an illustrative example. The rest of the sections are structured as follows: Section 2 details Impute-VSS's architecture, its functionalities and various characteristics. Section 3 showcases an example where we used a modified PIMA Indians diabetes dataset with missing values, to perform imputation using Impute-VSS. Section 4 discusses the potential impact and implications of Impute-VSS. Lastly, Section 5 delivers results which offer key takeaways, and also discusses the limitations of this research work, along with possible future directions.

2. Software description

2.1. Software architecture

The Impute-VSS is a centralized single-dashboard application, which comprises a front-end layer and a back-end layer. The communication between these layers is facilitated through AJAX calls, that aid in responsive and real-time interaction without performing hard reload to render. The front-end layer tries to offer an intuitive user interface, which is built on a single-page application (SPA) architecture. It is structured through the combination of HTML, CSS, JavaScript, and libraries such as Plotly.js for visualization of data in real-time. This layer allows users to upload datasets, select imputation methods, manage imputation pipelines and view statistical comparisons. The user interface of Impute-VSS is composed of a tri-partite menu structure which includes a *Navigation Bar, Imputation Playground* and (refer to Fig. 1 for a diagrammatic representation of Impute-VSS architecture). All these menus remain stationary during the operation, enabling users to interact with the features at all times. The back-end layer handles data preprocessing and computations of imputation. This layer is built using Python with Flask [11] as the primary framework, which leverages libraries like NumPy, Pandas, and SciPy for statistical operations. The back-end also manages the modular imputation pipeline which stores the user-specific configurations for reproducibility of the similar imputation workflow. The toolkit utilizes scikit-learn to process imputation requests on the backend, ensuring that the suite's performance is directly aligned with prevalent scikit-learn implementations.

Together, these layers aim to formulate a cohesive system, where the front-end layer communicates with the back-end asynchronously, thereby ensuring that imputation operations and visual updates occur concurrently without reloading the entire dashboard.

2.2. Software functionalities

Impute-VSS possesses numerous characteristics, which are sectionally presented below that explain the technical, aesthetical and mathematical underpinnings accordingly.

2.2.1. UI and UX characteristics

The user interface of Impute-VSS is designed to facilitate efficient imputation workflows by offering a clean and consistent user experience. The centralized dashboard is crafted using a single-tone blue hue palette, to establish a synchronized design language across the interface. The suite uses screen overlays to present menus and collapsible windows, such as the prompt modification menu, comparator menu and MVGenerator menu. The charts generated within the application are rendered using Plotly.js, which enables zooming, panning, and interactive tooltips on each data point, which helps in presenting crucial information.

2.2.2. Univariate and multivariate techniques

Impute-VSS offers multiple imputation techniques, which can be classified as univariate and multivariate imputation. The imputation of

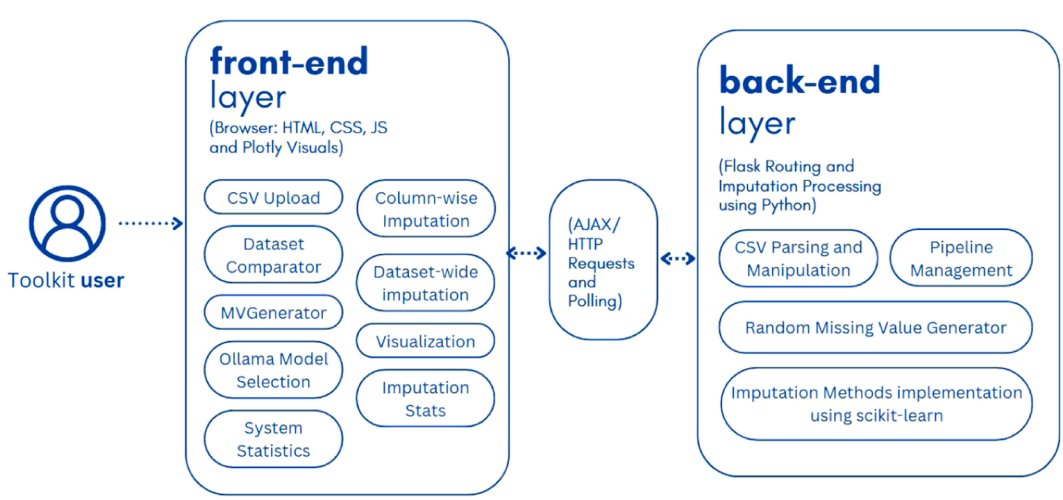


Fig. 1. Software Architecture of Impute-VSS.

dataset is computed using scikit-learn library. The Univariate Imputation Method addresses the missing data by independently estimating the absent values based on the observed entries of the variable under investigation [12]. For a numerical column, Impute-VSS offers various univariate imputation methods such as mean-value imputation, mode-value imputation, median-value imputation, random sampling imputation and constant-value imputation [13–15]. Also, Impute-VSS hosts machine learning algorithms such as Gradient Boosting, Random Forest Regressor, and Support Vector Regression for imputation of a column, while taking the whole dataset in the context [16–18]. In the case of the categorical column, it presents mode-value imputation, random-sampling imputation and constant-value imputation. The Univariate Imputation methods included in Impute-VSS can be mathematically expressed as follows (refer to Eqs. (1) to (5)):

$$Mean-value\ Imputation: \left(\widehat{x_{\text{missing}}} = \bar{x} = \frac{1}{n}\sum_{i=1}^{n} x_i\right) \tag{1}$$

$$Median-value\ Imputation: \left(\widehat{x_{\text{missing}}} = \text{median}\{x_1, x_2, \cdots, x_n\}\right) \tag{2}$$

$$Mode-value\ Imputation: \left(\widehat{x_{\text{missing}}} = \underset{x\in X}{\text{argmax}}\ \text{count}(x)\right) \tag{3}$$

$$Constant-value\ Imputation\ \left(\widehat{x_{\text{missing}}} = c\right) \tag{4}$$

$$\left(\widehat{x_{\text{missing}}} = \text{RandomSampling}\ (x_i | x_i \text{ is observed}\}\right) \tag{5}$$

In contrast to univariate, multivariate imputation techniques such as Multivariate Imputation by Chained Equations (MICE) and K-Nearest Neighbours (KNN) leverages entire context of variables in the dataset to impute the missing values [19,20]. In KNN, the Euclidean distance $((d(\boldsymbol{x}_i, \boldsymbol{x}_j))$ between observations i and j with common observed features is calculated, and the missing value is imputed as the average of the corresponding values from its k nearest neighbours [21], which can be mathematically expressed as (Refer to Eqs. (6) and (7)):

$$\left(d(x_i, x_j) = \sqrt{\sum_{l\in\mathscr{O}_{ij}} (x_{il} - x_{jl})^2}\right) \tag{6}$$

$$\left(\widehat{x_{\text{missing}}} = \frac{1}{k}\sum_{j\in\mathscr{N}_k(i)} x_j\right) \tag{7}$$

While MICE is an iterative, multivariate imputation method; that estimates missing data by modeling each variable with missing entries conditionally on all the other variables in the dataset [22]. For each variable, a regression model is built using the other variables (treated as predictors) to estimate the missing entries. This regression model, denoted as g_j for the jth variable, is then used to predict and fill in the missing values [23], so that for an observation i with missing data in $\widehat{x_{ij}}$, the imputed value is given by (Refer to Eq. (8), wherein p is the total number of variables):

$$\left(\widehat{x_{ij}} = g_j\left(x_{i1}, \cdots, x_{i(j-1)}, x_{i(j+1)}, \cdots, x_{ip}\right)\right) \tag{8}$$

Impute-VSS offers KNN and MICE on numerical as well as categorical variables. To handle the imputation of categorical variables within these multivariate frameworks, Impute-VSS applies label encoding which helps in mapping each unique category present in the variable X, to a unique numerical value [24]. This encoding and subsequent decoding facilitates the use of modeling techniques such as MICE and distance-based calculations in KNN.

In Impute-VSS, users can perform imputation step-by-step on each column, while concurrently assessing the parameters of initiated imputation techniques. Each univariate imputation method can be applied to the column individually during the analysis. This partially imputed dataset then acts as the reference dataset for further imputation. However, if the user applies a multivariate imputation from any column, the entire dataset must be imputed since multiple variables are interrelated. Thus, the user may perform a multivariate imputation on a column for comparative assessment alongside univariate imputation, but once any multivariate imputation strategy is initiated and applied through any column, the whole reference dataset is affected, thereby completing the imputation process. Apart from this, Impute-VSS offers Complete Case Analysis (CCA) as well, in which all the instances constituting missing values is eliminated. To showcase the comparative analysis of imputation techniques, Impute-VSS renders various plots such as a dataset-wide heatmap which showcases changes in the missing value trends across the dataset in a single glance. For the numerical column, toolkit generates a Kernel Density plot that shows the probability density function across the observed values, a histogram which visualizes frequency distribution through defined bins, a boxplot to display the inter-quartile range (IQR) with potential outliers, and a correlation line-chart which depicts the variation in correlation with other columns on performing imputation. Additionally, for statistically evaluating the imputation parameters, Impute-VSS renders a detailed statistics table. The toolkit can handle numerical, categorical and mixed-data types accordingly as each column under operation is systematically assessed.

For the numerical columns, this table include metrics such as mean, median, standard deviation, skewness, kurtosis, KDE overlap, and test statistics (e.g., Kolmogorov–Smirnov statistic and p-value, along with KL divergence), which collectively facilitate a comparison between the original and imputed data distributions. For categorical columns, the statistics table details measures such as the mode, unique count, mode frequency, and unique percentage. These visual and tabular outputs together enable practitioners to assess the efficacy and nuances of each imputation strategy; thereby, supporting informed decision-making. Impute-VSS provides a 'Comparator' feature that enables users to compare the actual dataset with the imputed dataset using evaluation metrics such as MAE, MSE, and RMSE. Additionally, it includes 'MVGenerator', a feature that artificially introduces missing values into dataset columns at random, based on user-defined percentages.

Table 2
Evaluation Metrics Comparison of Imputed 'BMI' using MICE with original dataset.

Percentage Missing Values	MAE	MSE	RMSE
20 %	0.9031	8.2249	2.8679
40 %	2.1097	20.8808	4.5695
60 %	3.156	29.3567	5.4182
80 %	4.1762	41.1747	6.4168

3. Illustrative examples

To demonstrate the functionality of Impute-VSS, this study formulated a modified PIMA Indians diabetes dataset. Since PIMA dataset has been widely used in multiple studies, and is aggregated by the National Institute of Diabetes and Digestive and Kidney Diseases (NIDDK) [25]. We generated four modified datasets using the MVGenerator feature in Impute-VSS, each incorporating missing values in 'BMI' variable at varying levels of 20 %, 40 %, 60 %, and 80 % (refer to Table 2 for evaluation metrics analysis with original dataset). To showcase the imputation and corresponding analysis using Impute-VSS, we opted to use the 'BMI' variable at 20 % missing values, in the modified PIMA Indians dataset through Mean, KNN, MICE and Median Imputation. In the analysis; KNN, MICE, Random Forest Regressor and Support Vector Machines algorithms were used to impute the BMI variable (refer to Fig. 3, and Fig. 4 for the comparative assessment). On examining the modified PIMA Indians diabetes dataset (20 % missing values), the evaluation metrics: MAE, MSE, and RMSE demonstrate a clear trend; as the percentage of missing values increases from 20 % to 80 %, the

```
OUTPUT    DEBUG CONSOLE    TERMINAL    PORTS

PS C:\Users\vartu\Downloads\ImputeVSS-16012025-main\ImputeVSS-16012025-main> & C:/Users/vartu/AppData/Local/Programs/Python/Python310/
S-16012025-main/ImputeVSS-16012025-main/app.py
 * Serving Flask app 'app'
 * Debug mode: on
WARNING: This is a development server. Do not use it in a production deployment. Use a production WSGI server instead.
 * Running on http://127.0.0.1:5000
Press CTRL+C to quit
 * Restarting with stat
 * Debugger is active!
 * Debugger PIN: 701-356-499
127.0.0.1 - - [26/Feb/2025 11:30:13] "GET /system_stats HTTP/1.1" 200 -
127.0.0.1 - - [26/Feb/2025 11:30:15] "GET / HTTP/1.1" 200 -
127.0.0.1 - - [26/Feb/2025 11:30:15] "GET /system_stats HTTP/1.1" 200 -
127.0.0.1 - - [26/Feb/2025 11:30:15] "GET /static/css/styles.css HTTP/1.1" 304 -
127.0.0.1 - - [26/Feb/2025 11:30:15] "GET /static/images/imputevss-front.png HTTP/1.1" 304 -
127.0.0.1 - - [26/Feb/2025 11:30:15] "GET /static/images/imputevss-logo.svg HTTP/1.1" 304 -
127.0.0.1 - - [26/Feb/2025 11:30:15] "GET /static/images/imputevss-desc.svg HTTP/1.1" 304 -
127.0.0.1 - - [26/Feb/2025 11:30:15] "GET /static/images/csv.png HTTP/1.1" 304 -
127.0.0.1 - - [26/Feb/2025 11:30:15] "GET /static/js/script.js HTTP/1.1" 304 -
127.0.0.1 - - [26/Feb/2025 11:30:15] "GET /static/images/impute-vss-bg.png HTTP/1.1" 304 -
127.0.0.1 - - [26/Feb/2025 11:30:16] "GET /system_stats HTTP/1.1" 200 -
127.0.0.1 - - [26/Feb/2025 11:30:19] "GET /system_stats HTTP/1.1" 200 -
```

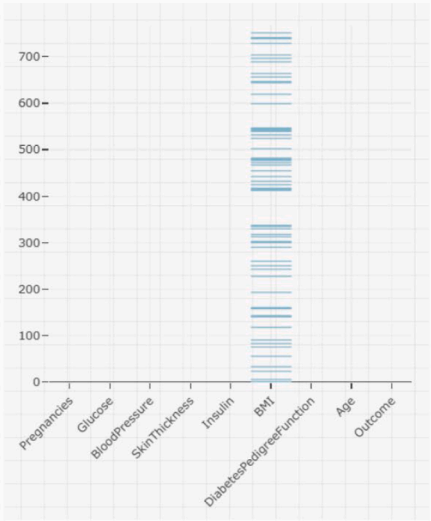
Fig. 2. System logs in the Python environment during the initialization of Impute-VSS.

imputation error also correspondingly increases (refer to Table 2). This trend indicates that with lower missing values, imputation method (MICE, in this case) can predict values more accurately, as stated by the lower MAE value of 0.9031, MSE of 8.2249, and RMSE of 2.8679 at 20 % missingness in 'BMI' variable. Conversely, as the missing data increases, the average error (MAE rising to 4.1762) and the impact of larger deviations (MSE increasing to 41.1747, with RMSE reaching 6.4168) are pronounced. This observation emphasizes upon the challenge of maintaining imputation quality when missing values becomes more prominent within the dataset.

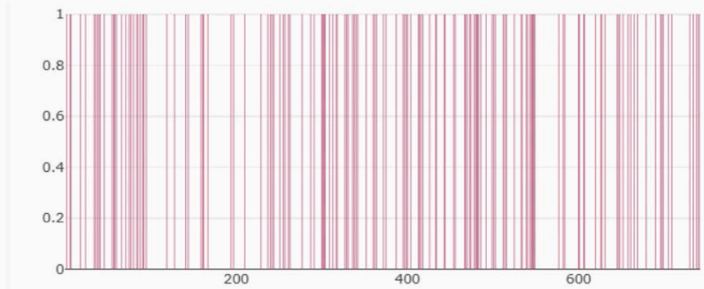
4. Impact

Through the development of Impute-VSS, this research aims to provide an inclusive suite for data imputation that can be employed across multiple domains. This toolkit aims to bridge the gap in the literature, by advancing an imputation toolkit that could be used by both experts and practitioners with limited familiarity with imputation methodologies.

Fig. 3. Comparison of imputation strategies in Impute-VSS Playground. It shows how different methods affect data distribution, missing values, and variable relationships using histograms, KDE plots, missing bars, box plots, and correlation plots to guide method selection.



Heatmap of Missing Values in Modified PIMA Dataset



Missing Values Bar Chart for variable BMI in Modified PIMA Dataset

Method	Time Taken	Mean	Median	Std	KDE Overlap	Skew	Kurt	KS Stat	KS p-value	KL Divergence
Original	—	31.93447	32.00000	8.04804	—	-0.46941	3.35231	—	—	—
knn	0.02909	32.07805	32.29000	7.45189	0.94353	-0.51598	4.02242	0.03651	0.73435	0.01039
mice	0.04763	32.06507	32.40000	7.41311	0.93724	-0.52683	4.14749	0.03844	0.67475	0.01276
random	0.02517	31.72474	32.00000	8.17465	0.98115	-0.61247	3.53032	0.01218	1.00000	0.00044
rf	0.51898	32.10629	32.29550	7.48561	0.94937	-0.49705	3.92033	0.03230	0.85311	0.00892
svr	0.04863	31.90812	31.95595	7.26213	0.90291	-0.50174	4.66858	0.05537	0.23441	0.02938

Table detailing comparative assessment of various imputation techniques

Fig. 4. Dataset-Wise Heatmap, Missing Bars Chart, and Statistical Table of Modified PIMA Indians Dataset (20% Missing in "BMI"). The heatmap and missing bars chart reveal missing data patterns, while the table details comparative assessment across imputation techniques for practitioner's reference.

Impute-VSS offers multiple univariate, machine learning and multivariate imputation techniques. Along with it, the suite concurrently offers visual diagnostic feedback through different charts, which enables granular insights. This comparative assessment through diverse techniques could help in addressing the domain and context-specific requirements of the practitioners. Thus, by integrating imputation methodologies with real-time visual feedback along with an accessible interface, Impute-VSS tries to deliver a significant impact by simplifying the imputation process, to enable informed decision-making across diverse research and industry applications.

5. Conclusions and future outlook

This study sheds light on the technical architecture of Impute-VSS, along with the potential impact it could bring in the field of data processing. To conclude, Impute-VSS aims to offer the following Key Features (KFs):

K.*F.1 To deliver a GUI-based interface that offers diverse univariate, machine learning and multivariate imputation techniques to render comparative reasoning while imputing the dataset.*

K.*F.2 To provide concurrent real-time visual feedback through missing values heatmap, Kernel Density Estimation (KDE), histogram, box plot, correlation line chart and missing values* bar *chart.*

K.*F.3 To showcase the statistical significance of each imputation technique applied to variables of the dataset through calculation of Kernel Density Estimation (KDE) overlap, Skewness (Skew), Kurtosis (Kurt), Kolmogorov-Smirnov Statistic (KS Stat), Kolmogorov-Smirnov p-value (KS p-value), and Kullback-Leibler Divergence (KL Divergence), along with standard metrics concerning the original distribution.*

K.*F.4 To ensure modularity, the Imputation Pipeline in Impute-VSS stores applied steps, enabling easy sharing and replication without repeatedly transmitting the dataset.*

However, there are several areas of improvement and future avenues which need to be acknowledged, that could strengthen the impact and applicability of Impute-VSS holistically. For instance, the incorporation of deep learning methodologies can help in drawing more profound comprehensive comparative assessments [26]. While Impute-VSS offers a web-based implementation to cater to cross-platform compatibility, similar toolkits can be separately developed for varying environments, that can analyse big data by utilizing multi-processing and multi-threading capabilities. Hence, by acknowledging these limitations and future avenues, Impute-VSS aims to position itself as a comprehensive web-based data imputation toolkit; that would help practitioners, researchers, developers and individuals with a vested interest in data imputation. By fostering innovation in imputation methodologies and expanding cross-disciplinary applications, Impute-VSS aims to catalyse a new wave of user-centric software development, ultimately empowering practitioners to navigate the complexities of data imputation with ease and precision.Fig. 2

CRediT authorship contribution statement

Vartul Shrivastava: Writing – original draft, Visualization, Validation, Software, Methodology, Data curation, Conceptualization. **Shekhar Shukla:** Writing – original draft, Validation, Supervision, Resources, Project administration, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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